

CSE 200 : Online - 1 on L^AT_EX (A1)

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Mark Distribution

Topic	Marks
Document Structure, Calligraphic Title & Two-column Layout	5
Lists, Text Formatting, Emphasis & Colors	4
Equations, Cases, Matrices & Mathematical Notation	14
Multirow and Multicolumn Table	6
Rotated Figure, Caption & Label	4
Algorithm and Complexity Expression	5
Python Code Listing and Syntax Formatting	6
Citations & Bibliography	2
Cross-references	2
Overall Presentation & Successful Compilation	2
Total	50

Knot Theory

Diagrams, Invariants, and Applications

Peter Guthrie Tait¹

¹Royal Society of Edinburgh, Scotland, UK

1 Knots as Mathematical Objects

1.1 From Rope to Curve

A mathematical knot is a closed curve in three-dimensional space that has no self-intersections. Formally, it may be represented by a continuous one-to-one map

$$\mathcal{K} : S^1 \longrightarrow \mathbb{R}^3, \quad (1)$$

where S^1 denotes the unit circle. Two knots are considered equivalent if one can be continuously deformed into the other without cutting the curve or allowing it to pass through itself [1].

The simplest knot is the *unknot*, which can be deformed into an ordinary circle. A nontrivial example is the trefoil knot shown in Figure 1.

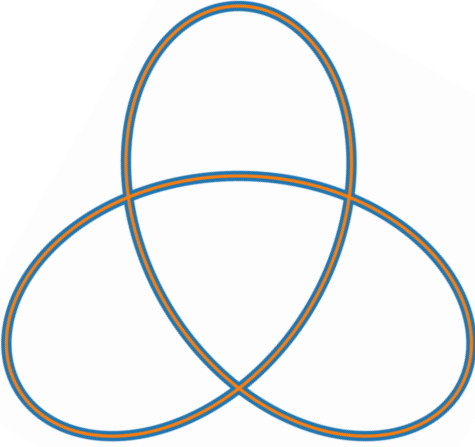


Figure 1: A planar drawing of a trefoil knot.

1.2 Diagrams and Crossings

A knot diagram is a projection of a spatial knot onto a plane. At every crossing, a small break indicates which strand passes underneath. The main diagrammatic objects are:

Arc	A continuous segment between two under-crossings.
Crossing	A point where one projected strand passes over another.
Orientation	A selected direction of travel around the knot.

A diagram may contain many crossings even when the underlying knot is simple, while an invariant must remain unchanged under every legal deformation of the knot.

2 Crossing Number and Writhe

2.1 Crossing Number

For a diagram D , let $c(D)$ be its number of crossings. The crossing number of a knot K is

$$c(K) = \min_{D \sim K} c(D), \quad (2)$$

where $D \sim K$ means that D represents the knot K . Thus,

$$c(\text{unknot}) = 0, \quad c(\text{trefoil}) = 3.$$

2.2 Signed Crossings

For an oriented diagram, each crossing is assigned a sign $\varepsilon_i \in \{+1, -1\}$. The writhe of the diagram is

$$w(D) = \sum_{i=1}^{c(D)} \varepsilon_i = n_+(D) - n_-(D), \quad (3)$$

where $n_+(D)$ and $n_-(D)$ count positive and negative crossings. For example, if the signs are $+1, +1, -1, +1$, then

$$w(D) = 1 + 1 - 1 + 1 = 2.$$

Table 1: Selected knots and several elementary invariants.

Family	Knot	Selected invariants			Chiral?
		$c(K)$	$\det(K)$	$\Delta_K(t)$	
Basic	Unknot	0	1	1	No
Torus knots	Trefoil 3_1	3	3	$t^{-1} - 1 + t$	Yes
	Cinquefoil 5_1	5	5	$t^{-2} - t^{-1} + 1 - t + t^2$	Yes
Twist knot	Figure-eight 4_1	4	5	$-t^{-1} + 3 - t$	No

Writhe is useful in calculations, but it is not itself a knot invariant: it can change under a Reidemeister move of type I.

3 Reidemeister Moves

Two diagrams represent equivalent knots exactly when one can be changed into the other by planar deformation together with the three Reidemeister moves [2]:

- (I) **Twist move:** adds or removes one loop and one crossing;
- (II) **Poke move:** adds or removes two neighboring crossings;
- (III) **Slide move:** moves one strand across a crossing.

These moves preserve the underlying knot type even though the visible number and arrangement of crossings may change.

4 Knot Invariants

4.1 Alexander Polynomial

A knot invariant assigns the same value to every diagram of the same knot. One important example is the Alexander polynomial $\Delta_K(t)$, defined up to multiplication by $\pm t^m$. A common normalization satisfies

$$\Delta_K(t) = \Delta_K(t^{-1}), \quad \Delta_K(1) = 1. \quad (4)$$

For the trefoil knot,

$$\Delta_{3_1}(t) = t^{-1} - 1 + t. \quad (5)$$

The determinant of a knot is obtained from the polynomial by

$$\det(K) = |\Delta_K(-1)|. \quad (6)$$

Hence,

$$\det(3_1) = |-1 - 1 - 1| = 3.$$

Table 1 compares these quantities for several small knots.

4.2 A Skein Relation

Many polynomial invariants are computed by comparing three diagrams that differ only inside a small region. For the Alexander polynomial, one form of the skein relation is

$$\Delta_{L_+}(t) - \Delta_{L_-}(t) = (t^{1/2} - t^{-1/2}) \Delta_{L_0}(t), \quad (7)$$

where L_+ , L_- , and L_0 denote a positive crossing, a negative crossing, and a smoothing, respectively.

4.3 The Signature of a Knot

Another useful knot invariant is the signature $\sigma(K)$, obtained from a Seifert matrix V . It is defined as the signature of the symmetric matrix

$$V + V^T,$$

that is, the number of positive eigenvalues minus the number of negative eigenvalues.

For a knot and its mirror image, the signature satisfies

$$\sigma(K) = \begin{cases} 0, & \text{if } K \text{ is the unknot,} \\ -2, & \text{if } K \text{ is the right-handed trefoil,} \\ 2, & \text{if } K \text{ is the left-handed trefoil.} \end{cases} \quad (8)$$

In particular, the signature distinguishes the right-handed trefoil from its mirror image, even though the two knots have the same Alexander polynomial.

5 A Small Diagram Algorithm

Algorithm 1 computes the writhe when the signed crossings of an oriented diagram are already known.

Algorithm 1 Compute the writhe of a knot diagram

Require: A list $S = (\varepsilon_1, \dots, \varepsilon_m)$ with $\varepsilon_i \in \{-1, +1\}$

Ensure: The writhe $w(D)$

```

1:  $w \leftarrow 0$ 
2: for each  $\varepsilon$  in  $S$  do
3:    $w \leftarrow w + \varepsilon$ 
4: end for
5: return  $w$ 
```

The algorithm performs one addition per crossing, so its running time is

$$T(m) = \Theta(m), \quad (9)$$

where $m = c(D)$.

6 A Short Python Implementation

Listing 1 implements the same calculation. It also checks that every crossing sign is valid.

```

1 def writhe(crossing_signs):
2     """Return the writhe of an oriented knot diagram."""
3     if any(
4         sign not in (-1, 1) for sign in crossing_signs
5     ):
6         raise ValueError("Each sign must be +1 or -1")
7     return sum(crossing_signs)
8
9
10 if __name__ == "__main__":
11     signs = [1, 1, -1, 1]
12     print(writhe(signs)) # Output: 2

```

Listing 1: Computing the writhe from signed crossings.

The mathematical definition in Equation (3) and the code in Listing 1 perform the same summation.

7 Applications Beyond Pure Mathematics

Knot theory appears whenever long flexible objects become entangled. Examples include:

- ⊕ circular DNA molecules and the action of topoisomerase enzymes;
- ⊕ polymer chains and molecular entanglement;
- ⊕ vortex lines in fluids and plasmas;
- ⊕ knot-based models in quantum computation.

For two oriented components C_1 and C_2 in a link diagram, the linking number can be calculated from crossings between the two components:

$$\text{lk}(C_1, C_2) = \frac{1}{2} \sum_{p \in C_1 \cap C_2} \varepsilon(p). \quad (10)$$

This quantity measures how many times the two components wind around one another, with orientation taken into account.

For a link $L = C_1 \cup \dots \cup C_n$, the pairwise linking numbers may be collected into the symmetric linking matrix

$$\Lambda(L) = [\lambda_{ij}]_{i,j=1}^n, \quad \lambda_{ij} = \begin{cases} \text{lk}(C_i, C_j), & i \neq j, \\ 0, & i = j. \end{cases} \quad (11)$$

References

- [1] C. C. Adams, *The Knot Book: An Elementary Introduction to the Mathematical Theory of Knots*, 2nd ed. American Mathematical Society, 2004.
- [2] L. H. Kauffman, *On Knots*. Princeton University Press, 1987.

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Topic	Marks
Document Structure, Title, Authors, and Sections	4
Text Formatting, Lists, Emphasis, and Colors	4
External Figure, Caption, and Label	6
Oscillator and Kuramoto Equations	6
Phase-Difference Derivation and Locking Condition	8
Table	6
Algorithm	6
Cross-references, Footnote, Hyperlink, Citations, and Bibliography	6
Overall Presentation and Successful Compilation	4
Total	50

Fireflies in Unison

How Simple Timing Rules Create a Collective Rhythm

Yoshiki Kuramoto¹ and Steven H. Strogatz²

¹Kyoto, Japan

²Ithaca, New York, USA

1 The Puzzle in the Forest

What do you think happens when hundreds of fireflies gather and begin flashing, each one following its own slightly different internal clock? At first, the scene may appear completely random. Yet, after observing and responding to nearby flashes, the individuals gradually adjust their timing until the entire group begins to pulse in a shared rhythm—with no leader, no conductor, and no central control [1]. That’s beautiful, right?

This is an example of *synchronization*: interacting systems that initially behave differently begin to follow a common rhythm. The same broad idea appears in pendulum clocks, cardiac cells, electrical generators, and communication networks.

The photograph in fig. 1 records light trails rather than a mathematical graph. The model below asks a simpler question:

What is the smallest timing rule that can turn many independent flash cycles into one coherent display?

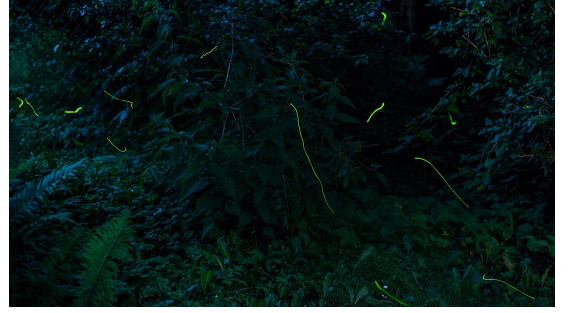


Figure 1: Firefly light trails in a garden. Photograph by Bernd Thaller, via Wikimedia Commons.

1.1 A Firefly as a Moving Clock Hand

Represent firefly i by a phase

$$\theta_i(t) \in [0, 2\pi).$$

One complete trip around the phase circle is one flash cycle. The firefly flashes when its phase reaches 2π , after which the phase begins again at 0. If it is isolated, then

$$\frac{d\theta_i}{dt} = \omega_i, \quad \theta_i(t) = (\theta_i(0) + \omega_i t) \bmod 2\pi, \quad (1)$$

where ω_i is its natural angular frequency. A larger ω_i means a shorter flash period,

$$T_i = \frac{2\pi}{\omega_i}. \quad (2)$$

Real fireflies emit brief pulses, so this continuous-clock picture is an approximation. ¹

2 Listening and Adjusting

2.1 The Kuramoto Rule

A widely used model for interacting phases is

$$\frac{d\theta_i}{dt} = \omega_i + \frac{K}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i), \quad i = 1, \dots, N. \quad (3)$$

The three parts of eq. (3) have direct meanings:

¹Pulse-coupled models represent each flash as a discrete event and are often closer to the biological mechanism [2].

Internal clock

ω_i is the rate the firefly would follow alone.

Coupling strength

$K \geq 0$ measures how strongly flashes influence one another.

Timing correction

$\sin(\theta_j - \theta_i)$ tells firefly i whether another phase is ahead or behind.

Table 1 explains the correction term. The table is simple, but it contains the central intuition of the model.

Table 1: How a neighboring firefly affects firefly i .

Relative position	Phase difference	Sine value	Effect on firefly i
Neighbor ahead	$0 < \theta_j - \theta_i < \pi$	Positive	Speeds up its phase.
Phases aligned	$\theta_j - \theta_i = 0$	Zero	Makes no correction.
Neighbor behind	$-\pi < \theta_j - \theta_i < 0$	Negative	Slows down its phase.
Opposite phases	$\theta_j - \theta_i = \pi$	Zero	Produces no preferred direction of correction.

The sine function is useful because it is smooth, periodic, and changes sign when the order of two phases is reversed.

3 Two Fireflies: An Exact Answer

The full model may contain hundreds of phases, but the case $N = 2$ can be solved directly. Define

$$\Delta = \theta_2 - \theta_1, \quad \Delta\omega = \omega_2 - \omega_1.$$

For two fireflies, eq. (3) becomes

$$\frac{d\theta_1}{dt} = \omega_1 + \frac{K}{2} \sin(\theta_2 - \theta_1), \quad (4)$$

$$\frac{d\theta_2}{dt} = \omega_2 + \frac{K}{2} \sin(\theta_1 - \theta_2). \quad (5)$$

Subtracting the first equation from the second gives

$$\frac{d\Delta}{dt} = \Delta\omega + \frac{K}{2} \sin(-\Delta) - \frac{K}{2} \sin(\Delta) \quad (6)$$

$$= \Delta\omega - K \sin \Delta. \quad (7)$$

The two fireflies are *phase locked* when Δ becomes constant. Setting the left side of eq. (7) to zero gives

$$\sin \Delta^* = \frac{\Delta\omega}{K}. \quad (8)$$

A real solution exists only when

$$\boxed{|\Delta\omega| \leq K}. \quad (9)$$

Thus, coupling can overcome a small mismatch between internal clocks, but not an arbitrarily large one.

4 From a Pair to a Population

4.1 Measuring Agreement

To measure how closely N phases are grouped, define the order parameter

$$r = \frac{1}{N} \sqrt{\left(\sum_{j=1}^N \cos \theta_j\right)^2 + \left(\sum_{j=1}^N \sin \theta_j\right)^2}, \quad 0 \leq r \leq 1. \quad (10)$$

Each phase is treated as a unit arrow. If the arrows point in many directions, they cancel and r is small. If they point together, they reinforce and r approaches 1.

As K increases, a typical population passes through three qualitative stages:

- I. **Drift:** phases move mainly according to their own natural frequencies.
- II. **Partial locking:** a group with similar frequencies begins to move together.
- III. **Strong coherence:** most phase differences remain bounded and the flashes form a stable rhythm.

5 A Short Simulation

Euler's method can approximate the phase evolution. The following compact algorithm repeats the interaction and update steps.

Algorithm 1 Simulate the Kuramoto model

Require: Phases θ_i , frequencies ω_i , coupling K , step size h , and number of steps M

```

1: for  $m \leftarrow 1$  to  $M$  do
2:   for each firefly  $i$  do
3:      $q_i \leftarrow \sum_{j=1}^N \sin(\theta_j - \theta_i), \quad v_i \leftarrow \omega_i + \frac{K}{N} q_i$ 
4:   end for
5:   Update all phases simultaneously:  $\theta_i \leftarrow (\theta_i + h v_i) \bmod 2\pi$ 
6: end for
7: return  $\theta_1, \dots, \theta_N$ 
```

Each firefly compares its phase with all N phases. Therefore, for M time steps,

$$T(M, N) = \Theta(MN^2). \quad (11)$$

6 What the Model Teaches

The Kuramoto model is deliberately small: it keeps only phase, natural frequency, and coupling. Yet it explains several general lessons.

- ◊ A shared rhythm can emerge without a leader or global clock.
- ◊ The same mathematical language can describe biological rhythms, power-grid generators, chemical oscillations, and distributed timing systems.

The model also has limitations. Real fireflies interact through visible pulses, have restricted fields of view, move through space, and may respond differently to males and females. Those details require richer pulse-coupled or spatial models. A field study and video of natural synchronous swarms are available through the [Science Advances study by Sarfati, Hayes, and Peleg](#).

References

- [1] Raphaël Sarfati, Julie C. Hayes, and Orit Peleg. “Self-Organization in Natural Swarms of *Photinus carolinus* Synchronous Fireflies”. In: *Science Advances* 7.28 (2021), eabg9259. DOI: [10.1126/sciadv.abg9259](https://doi.org/10.1126/sciadv.abg9259).
- [2] Renato E. Mirollo and Steven H. Strogatz. “Synchronization of Pulse-Coupled Biological Oscillators”. In: *SIAM Journal on Applied Mathematics* 50.6 (1990), pp. 1645–1662. DOI: [10.1137/0150098](https://doi.org/10.1137/0150098).

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Topic	Marks
Document Structure, Title, Authors & Affiliations	4
Lists, Nested Lists, Text Formatting & Colors	4
Laplace-transform Equations & Derivations	8
RC-circuit Equations, Alignments & Transfer Function	10
Multirow and Multicolumn Table	8
Wrapped and Rotated Figure, Caption & Label	4
Python Code Listing and Syntax Formatting	5
Cross-references	2
Footnote	1
Hyperlink	1
Citations & Bibliography	1
Overall Presentation & Successful Compilation	2
Total	50

Laplace Analysis and Applications

Pierre-Simon Laplace¹, Oliver Heaviside², and Gustav Robert Kirchhoff³

¹Académie des Sciences, Paris, France

²Royal Society, London, United Kingdom

³University of Heidelberg, Heidelberg, Germany

1 From Functions to Transforms

1.1 The Laplace Transform

Let $f(t)$ be a function defined for $t \geq 0$. Its Laplace transform is defined by

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^{\infty} e^{-st} f(t) dt, \quad s > 0. \quad (1)$$

The transform changes a function of time into a function of the new variable s .¹ Symbolically,

$$f(t) \xrightarrow{\mathcal{L}} F(s).$$

The inverse Laplace transform reverses this process:

$$f(t) = \mathcal{L}^{-1}\{F(s)\}. \quad (2)$$

The main purpose of the method is to replace differentiation and integration in the time domain by simpler algebraic operations in the transform domain [1]. A concise overview is available from the [Laplace Transform entry on MathWorld](#).

1.2 A Simple Example

For the constant function $f(t) = 1$,

$$\begin{aligned} \mathcal{L}\{1\} &= \int_0^{\infty} e^{-st} dt \\ &= \left[-\frac{1}{s} e^{-st} \right]_0^{\infty} \\ &= \frac{1}{s}. \end{aligned} \quad (3)$$

Similarly,

$$\mathcal{L}\{t\} = \frac{1}{s^2}, \quad \mathcal{L}\{t^2\} = \frac{2}{s^3}. \quad (4)$$

¹The variable s is commonly interpreted as a complex-frequency variable.

2 Common Transform Equations

Table 1 lists several frequently used Laplace-transform equations.

Table 1: Selected Laplace-transform equations.

Family	Transform Equation		Condition
	Time function	Laplace transform	
Algebraic	1	$\frac{1}{s}$	$s > 0$
	t	$\frac{1}{s^2}$	$s > 0$
	t^n	$\frac{n!}{s^{n+1}}$	$n = 0, 1, 2, \dots$
Exponential	e^{at}	$\frac{1}{s-a}$	$s > a$
	e^{-at}	$\frac{1}{s+a}$	$s > -a$
Oscillatory	$\sin(\omega t)$	$\frac{\omega}{s^2 + \omega^2}$	$\omega > 0$
	$\cos(\omega t)$	$\frac{s}{s^2 + \omega^2}$	$\omega > 0$
Unit-step function $u(t-a)$		$\frac{e^{-as}}{s}$	$a > 0$

The formulas in Table 1 allow many functions to be transformed without repeatedly evaluating improper integrals.

3 Application to an RC Circuit

Circuit Model

Figure 1 shows a simple resistor--capacitor circuit connected to an input voltage source. The resistor limits the current, while the capacitor stores electrical energy.

For a resistor R and capacitor C , Kirchhoff's voltage law gives

$$RC \frac{dv_C(t)}{dt} + v_C(t) = v_{in}(t), \quad (5)$$

where $v_C(t)$ is the voltage across the capacitor.

Assuming $v_C(0) = 0$, taking the Laplace transform of Equation (5) gives

$$\begin{aligned}
 RC [sV_C(s) - v_C(0)] + V_C(s) &= V_{in}(s), \\
 \Rightarrow RCsV_C(s) + V_C(s) &= V_{in}(s), \\
 \Rightarrow V_C(s)(RCs + 1) &= V_{in}(s).
 \end{aligned} \quad (6)$$

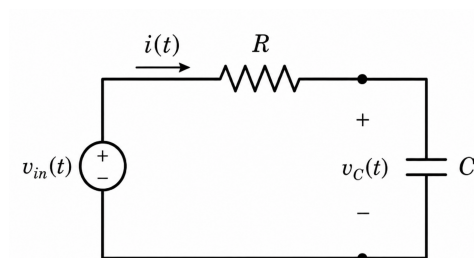


Figure 1: A simple series RC circuit.

The transfer function is therefore

$$H(s) = \frac{V_C(s)}{V_{\text{in}}(s)} = \frac{1}{RCs + 1}. \quad (7)$$

For a constant input V_0 ,

$$V_{\text{in}}(s) = \frac{V_0}{s}.$$

Thus,

$$V_C(s) = \frac{V_0}{s(RCs + 1)}. \quad (8)$$

Taking the inverse transform gives

$$v_C(t) = V_0 \left(1 - e^{-t/(RC)} \right). \quad (9)$$

The quantity $\tau = RC$ is called the time constant of the circuit. A larger value of τ causes the capacitor to charge more slowly [2].

4 How Laplace Analysis Is Used

The Laplace transform is useful in many areas:

Differential equations $\oplus$ initial conditions are included automatically;

$\oplus$ derivatives become algebraic expressions.

Electrical circuits $\oplus$ voltages and currents are studied in the s -domain;

$\oplus$ components lead to transfer functions.

Control and signal systems $\oplus$ stability can be studied through poles;

$\oplus$ common test inputs include:

▷ the unit-step input;

▷ the impulse input;

▷ sinusoidal inputs.

In the time domain, a system may be described by a differential equation, whereas in the Laplace domain, the same system is often described by a rational algebraic expression.

5 A Short Python Implementation

Listing 1 evaluates the capacitor voltage at a finite set of time instants using an external Python file.

```

1 from math import exp
2
3
4 def rc_response(resistance, capacitance, input_voltage, times):
5     """Return capacitor-voltage samples for an RC step input."""
6     if resistance <= 0 or capacitance <= 0:
7         raise ValueError("R and C must be positive")
8
9     tau = resistance * capacitance
10    values = []
11
12    for time in times:
13        voltage = input_voltage * (1 - exp(-time / tau))
14        values.append(voltage)
15
16    return values
17
18
19 times = [0.0, 0.5, 1.0, 2.0]
20 print(rc_response(2.0, 0.5, 5.0, times))

```

Listing 1: Computing samples of the RC step response.

Equation (9) and Listing 1 describe the same computation.

References

- [1] E. Kreyszig, *Advanced Engineering Mathematics*, 10th ed. John Wiley & Sons, 2011.
- [2] K. Ogata, *Modern Control Engineering*, 5th ed. Prentice Hall, 2010.

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Document Structure, Title, Authors, and Sections	4
Text Formatting, Lists, Emphasis, and Colors	4
External Figure, Caption, Label, and Cross-reference	6
Bit-to-Base Encoding, Equations, and Examples	8
DNA Strand Structure and Mathematical Notation	6
Two Tables and Sequence Formatting	12
Footnote, Hyperlink, Citations, and Bibliography	6
Overall Presentation and Successful Compilation	4
Total	50

A Library in a Droplet

How Digital Files Can Be Written into DNA

George M. Church¹ and Nick Goldman²

¹Harvard Medical School, Boston, USA

²European Bioinformatics Institute, Hinxton, UK

1 Can a File Become a Molecule?

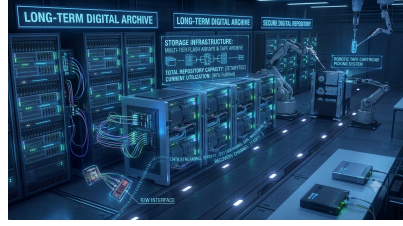
What would a library look like if its pages were replaced by molecules? Instead of shelves and hard drives, imagine a tiny tube containing millions of invisible DNA strands. The idea sounds like science fiction, yet researchers have already stored books, images, audio, software, and other digital files in synthetic DNA [1].

DNA normally carries biological instructions, but its four-letter alphabet can also carry arbitrary computer data. The DNA need not be placed inside a living organism: it may be synthesized, dried, stored, and later read by a sequencing machine. The central idea is simple:

Convert bits into DNA letters, preserve the molecules, and reverse the conversion when the file is needed again.



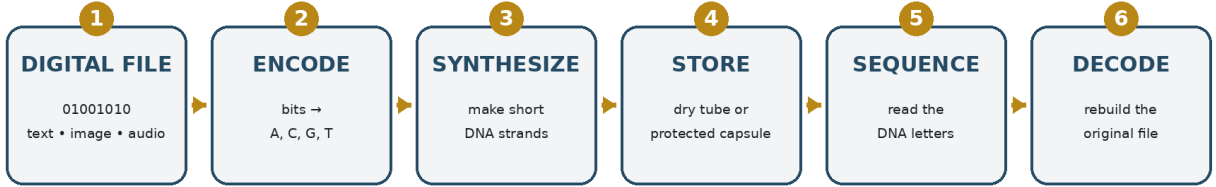
(a) DNA as a biological molecule.



(b) Digital data represented in bits.



(c) DNA used as a storage medium.



The write path goes left to right; reading reverses the process.

(d) A simplified DNA data-storage cycle, from a digital file to synthetic DNA and back again.

Figure 1: From biological DNA to a complete DNA-based digital storage system.

The six stages in fig. 1d separate the electronic world from the molecular one. The computer performs the encoding and decoding; laboratory instruments perform the writing and reading.

1.1 A Four-Letter Storage Medium

Ordinary digital files are sequences of bits, each equal to 0 or 1. DNA uses four symbols:

$$\mathcal{A} = \{A, C, G, T\}.$$

Because four symbols can represent four binary pairs,

$$4^n = (2^2)^n = 2^{2n}, \quad (1)$$

Table 1: One possible conversion from binary pairs to DNA bases.

Binary pair	DNA base
00	A
01	C
10	G
11	T

so an ideal DNA symbol can carry two bits of information.

Table 1 gives one possible toy mapping. Other mappings are equally valid.

For example, the character **A** has binary code 01000001 in ASCII.¹ Grouping the bits in pairs and using Table 1 gives

$$\mathbf{A} \longrightarrow 01\ 00\ 00\ 01 \longrightarrow \mathbf{CAAC}. \quad (2)$$

This example explains the principle, but a real DNA-storage system needs more structure than a direct translation.

2 Building a Molecular File

Long files are divided into many short DNA strands called *oligonucleotides*, or oligos. Each oligo contains not only data but also information that helps the system locate and reconstruct it.



If the corresponding lengths are L_p , L_a , L_d , and L_c , then the total strand length is

$$L = 2L_p + L_a + L_d + L_c. \quad (3)$$

The complete storage cycle can be summarized in five steps:

- I. Encoding:** convert a file into carefully designed DNA sequences.
- II. Synthesis:** manufacture those sequences as physical molecules.
- III. Preservation:** keep the DNA dry, cold, or chemically protected [2].
- IV. Sequencing:** read many copies of the stored strands.
- V. Reconstruction:** order the fragments, correct errors, and rebuild the original file.

Primer regions can also support *random access*: instead of reading an entire DNA pool, a user can request one particular file. In one large experiment, researchers stored more than 200 MB across 35 files and recovered each file individually [3].

3 Addresses, Payload, and Useful Capacity

Suppose a file is divided among M DNA strands. To assign a distinct address to every strand, the required address length is

$$L_a = \lceil \log_4 M \rceil, \quad (4)$$

because an address of L_a DNA bases can represent 4^{L_a} different positions.

¹ASCII is a standard numerical representation for common text characters.

Using the strand structure in eq. (3), the number of bases available for actual file data is

$$L_d = L - 2L_p - L_a - L_c. \quad (5)$$

Since one DNA base can ideally represent two bits, a file containing B bits requires approximately

$$M = \left\lceil \frac{B}{2L_d} \right\rceil \quad (6)$$

DNA strands. Thus, longer addresses and stronger error protection improve reliability, but leave less room for the data payload.

4 A Compact Encoding Procedure

The following procedure converts a bit string into addressed DNA strands. Here, $\text{Enc}(\cdot)$ applies the mapping in Table 1, and $\parallel$ denotes concatenation.

Algorithm 1 Create addressed DNA strands

Require: Bit string x , payload size q , and primers P_L, P_R

- 1: **for** each q -bit block x_i **do**
 - 2: $a_i \leftarrow$ base-4 address of i , $d_i \leftarrow \text{Enc}(x_i)$
 - 3: $c_i \leftarrow \text{Checksum}(a_i \parallel d_i)$
 - 4: $s_i \leftarrow P_L \parallel a_i \parallel d_i \parallel c_i \parallel P_R$
 - 5: **end for**
 - 6: **return** $\{s_1, s_2, \dots, s_M\}$
-

5 Why the Obvious Code Is Not Enough

The direct mapping in Table 1 is useful for learning, but biological hardware does not treat every DNA string equally well.

5.1 Sequence Constraints

Three common design concerns are:

- ◊ **Long repetitions:** strings such as AAAAAAA are difficult to synthesize and read accurately.
- ◊ **Unbalanced composition:** strands with extremely high or low proportions of G and C may behave poorly in laboratory procedures.
- ◊ **Lost ordering:** millions of strands are mixed together, so each fragment needs an address.

For a DNA string s , its GC percentage is

$$\text{GC}(s) = \frac{N_G(s) + N_C(s)}{|s|} \times 100\%, \quad (7)$$

where $N_G(s)$ and $N_C(s)$ count the two bases. Real encoders search for strings that carry the required bits while avoiding problematic patterns.

5.2 Errors at Different Stages

A stored strand can be altered in several ways:

Original	ACGTAC
Substitution	ACCTAC
Insertion	ACGTTAC
Deletion	ACTAC

Table 2 shows that reliability is not created by one single trick. Different stages require different safeguards.

Table 2: Where DNA-storage errors arise and how systems respond.

Phase	Stage	Challenge and response		
		Problem	Effect	Response
Writing	Encoding	Repeats or extreme GC content	Poor synthesis or reading	Choose another sequence
	Synthesis	Base errors	Incorrect strand	Add checks and copies
Preservation	Storage	Damage or strand loss	Missing fragments	Protect DNA and keep copies

6 Redundancy Without the Heavy Machinery

Error-correcting codes can become mathematically sophisticated, but their basic purpose is easy to see: store a little extra information so that imperfect reads do not destroy the message.

Suppose three sequencing reads report:

Read 1: ACGTAC
Read 2: ACGTAC
Read 3: ACCTAC

Recovered sequence: ACGTAC

A position-by-position majority vote repairs the single disagreement. Real systems combine this idea with checksums, overlapping fragments, and stronger codes. The DNA Fountain system, for example, stored about 2.14 MB of digital data and recovered it perfectly using a robust coding architecture [4].

The extra DNA is not wasted space; it is insurance against missing and corrupted molecules.

References

- [1] George M. Church, Yuan Gao, and Sriram Kosuri. “Next-Generation Digital Information Storage in DNA”. In: *Science* 337.6102 (2012), p. 1628. DOI: [10.1126/science.1226355](https://doi.org/10.1126/science.1226355).
- [2] Robert N. Grass et al. “Robust Chemical Preservation of Digital Information on DNA in Silica with Error-Correcting Codes”. In: *Angewandte Chemie International Edition* 54.8 (2015), pp. 2552–2555. DOI: [10.1002/anie.201411378](https://doi.org/10.1002/anie.201411378).
- [3] Lee Organick, Siena Dumas Ang, Yuan-Jyue Chen, et al. “Random Access in Large-Scale DNA Data Storage”. In: *Nature Biotechnology* 36 (2018), pp. 242–248. DOI: [10.1038/nbt.4079](https://doi.org/10.1038/nbt.4079).
- [4] Yaniv Erlich and Dina Zielinski. “DNA Fountain Enables a Robust and Efficient Storage Architecture”. In: *Science* 355.6328 (2017), pp. 950–954. DOI: [10.1126/science.aaj2038](https://doi.org/10.1126/science.aaj2038).

CSE 200: Online-1 on L^AT_EX (C1)

Time: 50 Minutes
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Mark Distribution

Topic	Marks
Document Structure, Title, Table of Contents & Two-column Layout	4
Text Formatting, Nested Lists, Colors & Emphasis	5
Equations, Alignments & Matrices	12
Colored Table with Multirow Formatting	7
Two-column Figure, Rotation, Caption & Label	4
Pseudocode and Complexity Expression	5
Theorem, Proof & Mathematical Environment	4
Cross-references	3
Footnote & Hyperlink	2
Citations & Bibliography	2
Overall Presentation & Successful Compilation	2
Total	50

The Riemann Hypothesis

Count Binface[‡]

[‡]Leader of Recyclon, Sigma IX & Master of L^AT_EX

Contents

1	A Function Built from Integers	3
1.1	The Riemann Zeta Function	3
1.2	A Classical Special Value	3
1.3	Two Mathematical Worlds	3
2	Zeros, Symmetry, and the Critical Line	3
2.1	Trivial and Nontrivial Zeros	3
2.2	The Functional Equation	3
2.3	The First Few Zeros	3
2.4	A Numerical Inspection Procedure	3
3	The Riemann Hypothesis	4
3.1	Formal Statement	4
3.2	Logical Interpretation	4
3.3	An Equivalent Error Statement	4
4	Why Prime Numbers Care	4
4.1	Mathematical Formulation	4

1 A Function Built from Integers

1.1 The Riemann Zeta Function

For a complex number $s = \sigma + it$ with $\sigma > 1$, the Riemann zeta function is

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_{p \text{ prime}} \left(1 - \frac{1}{p^s}\right)^{-1}. \quad (1)$$

The sum-product identity is the EULER PRODUCT and links $\zeta(s)$ with primes [1]. See the [Clay Mathematics Institute](#) [1].

1.2 A Classical Special Value

At $s = 2$, Equation (1) gives the classical identity

$$\zeta(2) = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \cdots = \frac{\pi^2}{6},$$

$$\prod_{p \in \{2,3,5,7\}} \left(1 - \frac{1}{p^2}\right)^{-1} \approx 1.595 < \frac{\pi^2}{6} \approx 1.645.$$

More prime factors improve the approximation.

1.3 Two Mathematical Worlds

Figure 1 connects zeta zeros with prime distribution.

2 Zeros, Symmetry, and the Critical Line

2.1 Trivial and Nontrivial Zeros

The *trivial zeros* are

$$-2, -4, -6, \dots$$

and the *non-trivial zeros* lie in the *critical strip*

$$0 < \text{Re}(s) < 1.$$

(A) Trivial zeros

- † They occur at negative even integers.
- † Their locations are completely understood.

(B) Nontrivial zeros

- ⊃ They lie inside the critical strip.
- ⊃ They occur in symmetric families.
- ⊃ Their symmetries generate the following companion points:
 - (i) ρ ;
 - (ii) $1 - \rho$;
 - (iii) $\bar{\rho}$;
 - (iv) $1 - \bar{\rho}$.
- ⊃ Their horizontal locations are central to the Riemann Hypothesis.

2.2 The Functional Equation

A symmetric form of the zeta function is

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s), \quad \xi(s) = \xi(1-s). \quad (2)$$

Equation (2) reflects zeros across $\sigma = 1/2$:

$$\begin{bmatrix} \sigma' \\ t' \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \sigma \\ t \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

The linear part of the reflection is the matrix

$$R = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, \quad R^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2. \quad (3)$$

Hence R is an involution.

2.3 The First Few Zeros

The line $\text{Re}(s) = \frac{1}{2}$ is the *critical line*.

Table 1: The first four positive ordinates of nontrivial zeros.

Index	γ_n	Real Part	Imaginary Part
1	14.134725	$\frac{1}{2}$	14.134725
2	21.022040		21.022040
3	25.010858		25.010858
4	30.424876		30.424876

The first two zeros form the matrix

$$Z_2 = \begin{bmatrix} \frac{1}{2} & 14.134725 \\ \frac{1}{2} & 21.022040 \end{bmatrix}, \quad (4)$$

Each row stores one zero.

2.4 A Numerical Inspection Procedure

Algorithm 1 tests candidate real parts.

Algorithm 1 Check candidate zeros against the critical line

Require: Candidate zeros $Z = (\rho_1, \rho_2, \dots, \rho_m)$ and tolerance $\varepsilon > 0$

Ensure: TRUE if every candidate satisfies

```

 $|\text{Re}(\rho_i) - \frac{1}{2}| \leq \varepsilon$ 
1: allOnLine  $\leftarrow$  TRUE
2: for  $i \leftarrow 1$  to  $m$  do
3:   if  $|\text{Re}(\rho_i) - \frac{1}{2}| > \varepsilon$  then
4:     allOnLine  $\leftarrow$  FALSE
5:   break
6: end if
7: end for
8: return allOnLine

```

Its running time is

$$T(m) = \Theta(m). \quad (5)$$

¹The Riemann Hypothesis is a Millennium Prize Problem.

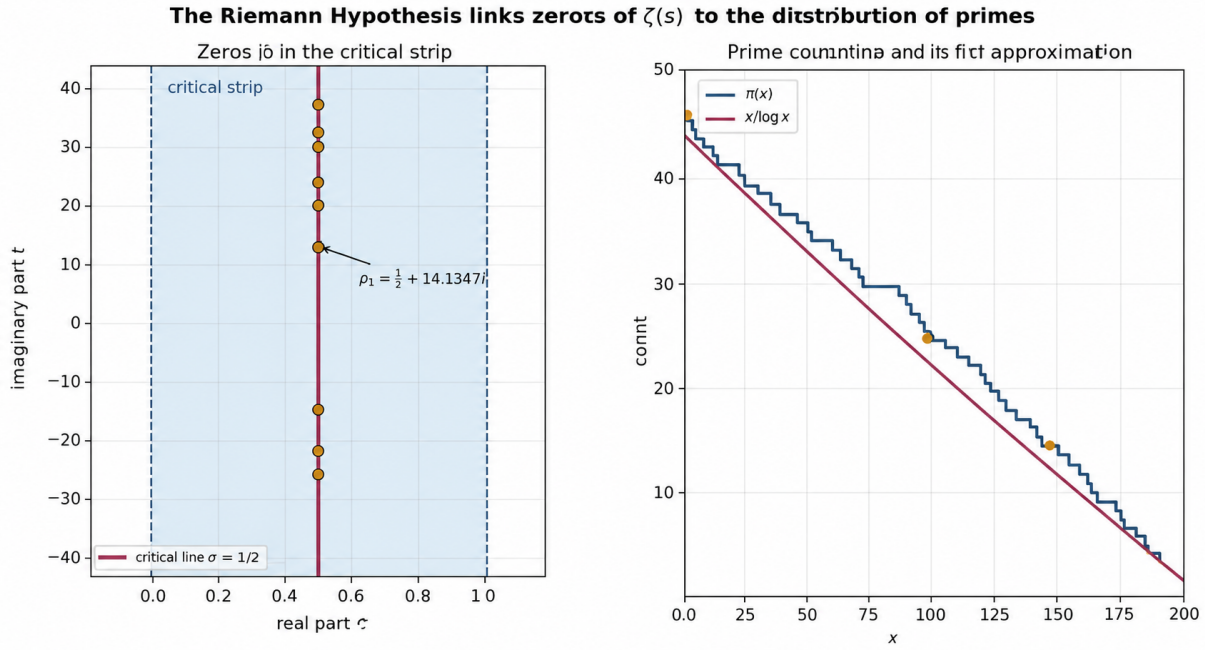


Figure 1: The critical strip and the relation between prime counting and analytic approximation.

3 The Riemann Hypothesis

3.1 Formal Statement

The Riemann Hypothesis states

$$\zeta(\rho) = 0, \quad 0 < \operatorname{Re}(\rho) < 1 \implies \operatorname{Re}(\rho) = \frac{1}{2}. \quad (6)$$

Equation (6) remains unproved [2].

3.2 Logical Interpretation

It asserts necessity, not sufficiency.

Thus,

$$\zeta(\rho) = 0 \implies \operatorname{Re}(\rho) = \frac{1}{2}$$

for nontrivial zeros; the converse fails in general.

3.3 An Equivalent Error Statement

Define the Chebyshev function

$$\psi(x) = \sum_{p^k \leq x} \log p.$$

Theorem 3.1 (Prime-distribution error bound). *The Riemann Hypothesis is equivalent to*

$$\psi(x) = x + O(\sqrt{x} \log^2 x). \quad (7)$$

Proof. The explicit formula expresses the error through the nontrivial zeros. The stated bound is equivalent to all such zeros having real part $\frac{1}{2}$. $\square$

Theorem 3.1 links zeros and prime-counting error.

4 Why Prime Numbers Care

Primes appear irregularly, but zeta zeros control that irregularity. RH sharply limits the prime-counting error.

4.1 Mathematical Formulation

Let $\pi(x) = \#\{p \leq x : p \text{ is prime}\}$ denote the prime-counting function. The prime number theorem states that

$$\pi(x) \sim \frac{x}{\log x}, \quad \lim_{x \rightarrow \infty} \frac{\pi(x) \log x}{x} = 1. \quad (8)$$

A sharper approximation is

$$\operatorname{Li}(x) = \int_2^x \frac{dt}{\log t}. \quad (9)$$

Under RH,

$$\pi(x) = \operatorname{Li}(x) + O(\sqrt{x} \log x). \quad (10)$$

Hence zeta zeros govern prime-counting fluctuations.

References

- [1] B. Riemann, “On the number of primes less than a given magnitude,” *Monatsberichte der Berliner Akademie*, 1859. English translation of the original German paper.
- [2] H. M. Edwards, *Riemann’s Zeta Function*. Academic Press, 1974.

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Mark Distribution

Topic	Marks
Document Structure, Title & Table of Contents	4
Lists, Text Formatting & Colors	4
Figure & Nine Subfigures	10
Keplerian and Planetary Equations	10
General Relativity Equations & Notation	8
Complex Colored Table	6
Proposition, Proof & Cross-references	3
References & Citations	1
Footnotes	1
Hyperlink	1
Overall Presentation & Successful Compilation	2
Total	50

THE SOLAR SYSTEM

Johannes Kepler¹

¹Imperial Mathematician, Prague

Contents

1 Our Planetary Neighborhood	3
1.1 The Sun and Its Planets	3
1.2 Portraits of the Solar System	3
2 Kepler's Laws of Planetary Motion	3
2.1 First Law: Elliptical Orbits	3
2.2 Second Law: Equal Areas in Equal Times	4
2.3 Third Law: Orbital Period and Distance	5
2.4 An Earth-Mars Comparison	5
3 A Relativistic Extension	5
4 The Outer Planets	6

1 Our Planetary Neighborhood

1.1 The Sun and Its Planets

The Sun and its gravity-bound bodies form the Solar System [1]. See [NASA Solar System Exploration]. In increasing distance, the planets are

Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune.

The planets form two groups:

(I) Inner planets

- † Mercury, Venus, Earth, and Mars belong to this group.
- † They are comparatively small and have solid surfaces.
- † They are also called the terrestrial planets.

(II) Outer planets

- † Jupiter and Saturn are gas giants.
- † Uranus and Neptune are commonly called ice giants.
- † All four outer planets possess ring systems.

The inner planets are small, dense, and relatively close to the Sun, whereas the outer planets are much larger and travel along considerably wider orbits.

1.2 Portraits of the Solar System

Figure 1 shows the Sun and eight planets.¹

Compare Mars in Figure 1c with Jupiter in Figure 1f

2 Kepler's Laws of Planetary Motion

Kepler's three laws describe planetary motion [2].

2.1 First Law: Elliptical Orbits

A planetary orbit is elliptical, with the Sun at one focus:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \quad a \geq b > 0, \quad (1)$$

Here a and b are the semi-axes.

¹Sizes and distances are schematic.

Star



(a) The Sun

Inner Planets



(b) Mercury



(c) Venus



(d) Earth



(e) Mars

Outer Planets



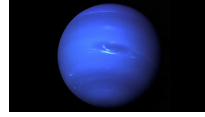
(f) Jupiter



(g) Saturn



(h) Uranus



(i) Neptune

Figure 1: The Sun and the eight planets of the Solar System, arranged into the inner and outer planetary groups. The displayed sizes and distances are not to scale.

For focal distance c ,

$$\begin{aligned} c^2 &= a^2 - b^2 \\ e &= \frac{c}{a} \end{aligned} \tag{2}$$

Here e is the eccentricity.

In polar form,

$$r(\theta) = \frac{a(1 - e^2)}{1 + e \cos \theta}. \tag{3}$$

Equation [\(3\)](#) gives the orbital distance.

2.2 Second Law: Equal Areas in Equal Times

Equal times sweep equal areas:

$$\Delta t_1 = \Delta t_2 \implies \Delta A_1 = \Delta A_2. \tag{4}$$

Consequently, a planet moves

- ◊ faster when it is closer to the Sun, and
- ◊ slower when it is farther from the Sun.

2.3 Third Law: Orbital Period and Distance

For planets orbiting one star,

$$T^2 \propto a^3. \quad (5)$$

Newtonian form:

$$T^2 = \frac{4\pi^2}{GM_\odot} a^3,$$

Here $M_\odot$ is the solar mass.

2.4 An Earth–Mars Comparison

Using Earth as reference,

$$a_E = 1 \text{ AU} \qquad T_E = 1 \text{ year}$$

where $1 \text{ AU} \approx 1.496 \times 10^{11} \text{ m}$.^[2] For Mars, $a_M \approx 1.524 \text{ AU}$, so

$$\begin{aligned} \frac{T_M^2}{T_E^2} &= \frac{a_M^3}{a_E^3}, \\ T_M &= T_E \left(\frac{a_M}{a_E} \right)^{3/2} \\ &\approx (1.524)^{3/2} \approx 1.88 \text{ years}. \end{aligned}$$

Thus $T_M \approx 1.88$ years.

3 A Relativistic Extension

General relativity extends Kepler's model.^[3] Einstein's equations are

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}. \quad (6)$$

For spherical mass M ,

$$\begin{aligned} ds^2 &= - \left(1 - \frac{r_s}{r} \right) c^2 dt^2 + \left(1 - \frac{r_s}{r} \right)^{-1} dr^2 \\ &\quad + r^2 (d\theta^2 + \sin^2 \theta d\phi^2), \quad r_s = \frac{2GM}{c^2}. \end{aligned} \quad (7)$$

²AU denotes the mean Earth–Sun distance.

³Planetary corrections are small but measurable.

Free fall satisfies

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} = 0. \quad (8)$$

Per orbit,

$$\Delta\varpi = \frac{6\pi GM}{a(1-e^2)c^2}. \quad (9)$$

Proposition 3.1 (Weak-field circular orbit). *For a circular orbit of radius r ,*

$$\left(\frac{v}{c}\right)^2 = \frac{GM}{rc^2} = \frac{r_s}{2r}. \quad (10)$$

Proof. The circular-orbit balance $v^2/r = GM/r^2$ gives $v^2 = GM/r$. Divide by c^2 and use $r_s = 2GM/c^2$. $\square$

See Proposition [3.1](#).

4 The Outer Planets

The outer planets are Jupiter, Saturn, Uranus, and Neptune [\[3\]](#). The first two are *gas giants*; the last two are *ice giants*.

Table [1](#) uses scientific and relativistic notation.[4](#)

Table 1: Orbital and relativistic parameters of the outer planets

Planet	Orbital parameters		Relativistic notation	
	a (AU)	e	$\varepsilon = \frac{1}{ac^2}GM_\odot$	$\beta = \sqrt{\varepsilon}$
Jupiter	5.204	4.89×10^{-2}	1.90×10^{-9}	4.36×10^{-5}
Saturn	9.583	5.65×10^{-2}	1.03×10^{-9}	3.21×10^{-5}
Uranus	19.191	4.72×10^{-2}	5.14×10^{-10}	2.27×10^{-5}
Neptune	30.070	8.6×10^{-3}	3.28×10^{-10}	1.81×10^{-5}

Table [1](#) compares the four giant planets.

References

- [1] J. Kepler, *Astronomia Nova*. Heidelberg, 1609.
- [2] National Aeronautics and Space Administration, “Orbits and kepler’s laws,” 2024.
- [3] National Aeronautics and Space Administration, “About the planets,” 2026. Solar System Exploration.

⁴Values are rounded for typesetting practice.